

RESCOPED PRODUCT STRATEGY

FPO-to-Kirana Direct B2B Produce Marketplace

A focused 6-week core transaction engine powered by Hindi voice-intake and automated Mandi price benchmarking.

1 Persona Pair

FPO Sellers & Kirana Buyers

100% Focused

Voice-to-Order B2B Flow



Radical Scope Reduction: B2B Bulk Focus

Dropped Secondary Features

- ✗ **CV Produce Grading:** Eliminated computer vision model overhead.
- ✗ **B2C Urban Consumers:** Dropped 1-hour doorstep delivery.
- ✗ **OR-Tools Heuristics:** Replaced with direct point pickup.

Retained Core B2B Scope

- ✓ **FPO Voice Listing:** Speech-to-text Hindi crop parsing.
- ✓ **Mandi Price Bounds:** Real-time dynamic price suggestion.
- ✓ **Kirana Bulk Orders:** Instant B2B REST order placement.

Core Value Metric: Enables 1 FPO to supply 50+ local Kirana stores with verified bulk pricing.

Voice-Intake & Listing Screen Spec

New Crop Listing

Hindi Voice

Tap & Speak

"कल 200 किलो टमाटर ready है"

Crop: Tomato (टमाटर)

Quantity: 200 Kg

Mandi Bound: ₹22 - ₹26 / kg

Publish Listing

- 

Whisper Speech Parser

Extracts crop entity, weight unit, and availability date in under 1.8 seconds.
- 

Dynamic Price Guidance

Fetches nearest Agmarknet Mandi rates to prevent under-pricing by small FPOs.
- 

Human-in-the-Loop Override

FPO leads maintain 100% manual control to adjust price before final posting.

Core REST Endpoints & JSON Schemas

POST

`/api/v1/listings/voice`

```
// Input Audio Payload & Output JSON
```

```
{
  "fpo_id": "fpo_8921",
  "audio_base64": "UklGRi...",
  "language": "hi"
}
```

```
// Response (200 OK)
```

```
{
  "crop": "Tomato",
  "qty_kg": 200,
  "benchmark_min": 22.0,
  "benchmark_max": 26.0
}
```

POST

`/api/v1/orders/create`

```
// Kirana Order Payload
```

```
{
  "kirana_id": "store_441",
  "listing_id": "lst_9902",
  "order_qty_kg": 50,
  "agreed_price": 24.0
}
```

```
// Response (201 Created)
```

```
{
  "order_id": "ord_1102",
  "status": "CONFIRMED",
  "pickup_slot": "2026-09-29T08:00:00Z"
}
```

PostgreSQL Data Model & Flow

PostgreSQL Relational Tables

users

id (PK), role (FPO/Kirana), phone, location_point

listings

id (PK), fpo_id (FK), crop_name, total_qty_kg, price_per_kg

orders

id (PK), listing_id (FK), kirana_id (FK), order_qty, status

1 Audio Ingestion & NLP
Whisper transcribes audio; regex & NER extract quantity and crop type.

2 Mandi Price Benchmark
Queries local Agmarknet pricing cache stored in Redis memory.

3 Order Creation & Stock Lock
Atomic SQL transaction decreases available quantity and emits order event.

Focused 6–Week Build Plan

Weeks 1 - 2

Voice & Database

- FastAPI & PostgreSQL Setup
- OpenAI Whisper Hindi Integration
- Crop Entity Extraction Pipeline

Weeks 3 - 4

B2B Marketplace CRUD

- Listing & Order Endpoints
- Agmarknet API Price Ingestion
- React Native Kirana Screen

Weeks 5 - 6

Testing & Pilot Deployment

- End-to-End Latency Optimization
- Pilot with 1 FPO & 10 Kirana Stores
- OpenAPI Docs Finalization

Image Sources



<https://fsii.in/wp-content/uploads/2022/08/Farmer-1.png>

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